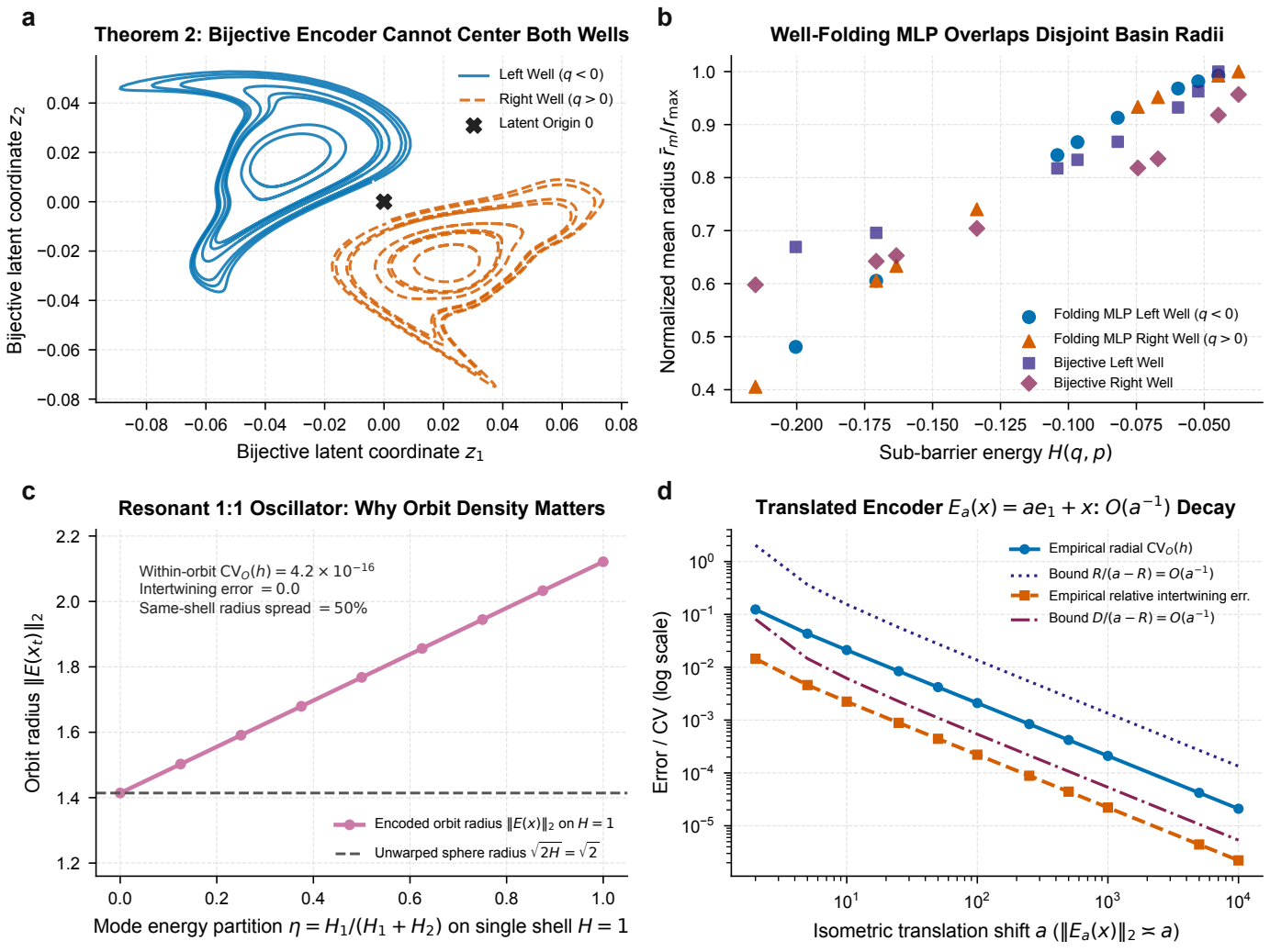

a
Theorem 2: Bijective Encoder Cannot Center Both Wells
Left Well ($q < 0$)
Right Well ($q > 0$)
Latent Origin 0
Bijective latent coordinate z_2
Bijective latent coordinate z_1
b
Well-Folding MLP Overlaps Disjoint Basin Radii
Folding MLP Left Well ($q < 0$)
Folding MLP Right Well ($q > 0$)
Bijective Left Well
Bijective Right Well
Normalized mean radius $\bar{r}_m / r_{\max}$
Sub-barrier energy $H(q, p)$
c
Resonant 1:1 Oscillator: Why Orbit Density Matters
Within-orbit $\mathrm{CV}_O(h) = 4.2 \times 10^{-16}$
Intertwining error = 0.0
Same-shell radius spread = 50%
Encoded orbit radius $\|E(x)\|_2$ on $H = 1$
Unwarped sphere radius $\sqrt{2H} = \sqrt{2}$
Orbit radius $\|E(x_t)\|_2$
Mode energy partition $\eta = H_1/(H_1 + H_2)$ on single shell $H = 1$
d
Translated Encoder $E_a(x) = ae_1 + x$: $O(a^{-1})$ Decay
Empirical radial $\mathrm{CV}_O(h)$
Bound $R/(a - R) = O(a^{-1})$
Empirical relative intertwining err.
Bound $D/(a - R) = O(a^{-1})$
Error / CV (log scale)
Isometric translation shift a ($\|E_a(x)\|_2 \asymp a$)